# PATIENT COMPLEXITY AND PATIENT NEEDS

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Patient complexity might be an intuitive concept to a primary care clinician, but is difficult to define in an agreed or precise manner [1]. The focus of complexity is sometimes an arbitrary - if significant - outcome such as risk of hospital admission [2, 3] based largely on 'medical' risk-factors. For a clinician, complexity may be experienced as challenging, multi-faceted, time-consuming and difficult to conduct consultations strongly influenced by psychological, socio-economic and systemic factors [4, 5]. Complexity can arise as the result of - sometimes problematic - interactions between patient, multiple health care providers and community [1]. Some define complexity as the gap between an individual's needs and the capacity of the healthcare system to answer those needs [6]. Broadly, complexity is associated with overall higher healthcare resource consumption and need for social support [1].

Patient complexity measurement tools are available in general practice electronic medical record (EMR) reporting tools which report practice population information to the Primary Health Networks (PHNs). PHN performance and quality framework program indicator 'P12' requires a focus on rate of potentially preventable hospitalisation [7].

- PenCS uses a CSIRO developed tool, designed to be used in the primary care context to predict the risk of hospitalisation using information available from Australian general practice EMRs. Tool workings and development, using logistic regression, is very well documented [8]. The tool can be fully implemented using the information provided. The authors state they are happy to share the methodology *exactly* for non-profit purposes, it may be possible to adapt the tool to other outcomes.
- POLAR uses the Diversion 2 tool, developed using machine learning techniques with demographic, diagnoses (coded and 'free text'), medication, pathology and measurement data extracted from general practices with GRHANITE and linked with hospital admission data [9, 10].
- Primary Sense uses the proprietary John Hopkins ACG System, which has been extensively developed and researched [11, 12]. The John Hopkins ACG system has been used to estimate current and likely future care needs in primary care, including more equitable funding of general practices in the NHS [13, 14]. However, the use of John Hopkins ACG in Primary Sense is limited to the risk of hospital admission. John Hopkins ACG uses *patient segmentation* to group patients into relatively homogenous groups with similar care needs based on medical and socio-economic characteristics [15–17] (Figure 1).

Opportunities

- Patient segmentation - with social need and other modifiers for special care needs - has potential value predict and provide care needs in primary care settings with tailoring towards behavioural, clinical or demographic drivers of health over and above predictors based on a single score [15, 17–19] (Figure 2). The concept has already generalized to a South-East Asian context [20] (Figures 3, 4).
- Predictive models for specific outcomes can be developed within patient segments, as both patient needs and importance of specific risk-factors varies between segments [21].
- Other intermediate outcomes can be explored e.g. frequency of incoming and outgoing correspondence to other healthcare providers.

# Figures

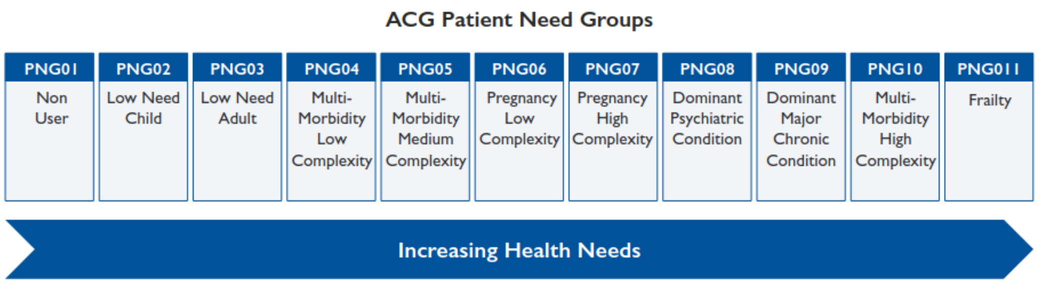


Figure 1: John Hopkins ACG Patient Needs Groups

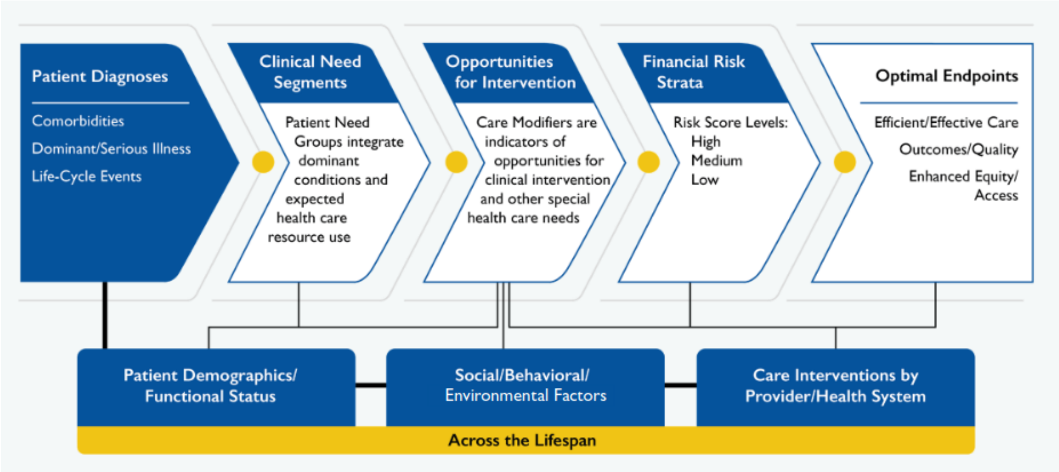


Figure 2: John Hopkins ACG care modifiers

<table>
<tr><th>Broad Groups</th><th>Children & Adolescence<br>18 years and below</th><th>Adults<br>19 - 64 years</th><th>Seniors<br>65 years and above</th></tr>
<tr><td rowspan="2">Staying Healthy<br>No Major Health Needs<br>(Preventive Care)</td><td colspan="3">[A] Mostly Healthy Children, Adults and Seniors<br>With no known chronic condition and no known clinical and social risk factor.<br>Maintenance of healthy lifestyle, age appropriate screening, age appropriate preventive health services (e.g. vaccination).</td></tr>
<tr><td colspan="2">[B] At Risk Children and Adults<br>With no known chronic condition and with one clinical or social risk factor such as low socio-economic status and obesity)<br>Targeted screening of at risk population (e.g. genetic/familial diseases), targeted interventions for at risk population.</td><td>[C] At Risk Seniors<br>With no chronic condition and one clinical or social risk factor.<br>Interventions to address risk factors to prevent disease.</td></tr>
<tr><td rowspan="4">Living Well with Illness<br>With Chronic Disease<br>(Chronic Disease Management)</td><td rowspan="4">[D] Children with Chronic Condition<br>Individuals who have chronic condition and require minimal follow-up at acute care.</td><td colspan="2">[E] Adults and Seniors with Simple Chronic Condition<br>With condition(s) and case-mix which can be managed primarily in primary care with minimal support from specialist care.</td></tr>
<tr><td colspan="2">[F] Adults and Seniors with Complex Chronic Condition<br>With condition(s) and case-mix which require appropriate co-management between primary and specialist care for chronic disease management.</td></tr>
<tr><td colspan="2">[G] Adults and Seniors with Mental Health Condition<br>With mental health condition. Require primary and specialist care for mental health, with support in the community.</td></tr>
<tr><td colspan="2">[H] Adults and Seniors with Cancer<br>With cancer. Require specialist care with support in the community.</td></tr>
<tr><td>Getting Well<br>Time-Limited Health Needs<br>(Transitional Care)</td><td rowspan="2">[I] Children with Transitional & Long-Term Needs<br>Require additional specialist care to return to baseline function and/or delay progression of disease.</td><td colspan="2">[J] Adults and Seniors with Transitional Needs<br>Had major acute episodic health needs and subsequently require additional support post-discharge and integrate back to the community and return of baseline function.</td></tr>
<tr><td rowspan="2">Maximising Quality Life Years<br>Frail and End-of-Life<br>(Disease Management and Palliative Care)</td><td colspan="2">[K] Frail Adults and Seniors<br>With diminished function (physical and/or cognitive) or are institutionalised in nursing homes.</td></tr>
<tr><td colspan="3">[L] End-of-Life Children, Adults and Seniors<br>Advanced care planning and accessible end-of-life services to support end-of-life preferences.</td></tr>
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Figure 3: Health Segment Classification of Population Encompassed within Singapore (HealthSCOPES)

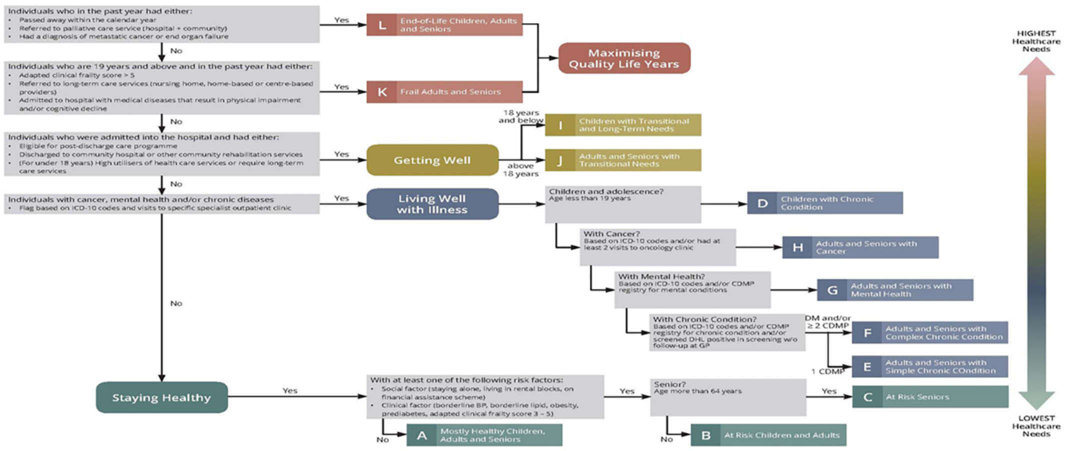


Figure 4: Classification hierarchy of HealthSCOPES, outlining the criteria to categorize an individual into one of the 12 segments

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